Response to Reviewers

IJADS-312370 — "From Signal Fusion to Asset Allocation: A Decision-Theoretic Framework for Regime-Conditional Portfolio Construction"

Summary of the revision

We thank both reviewers and the Editor for a careful and demanding review. Every mandatory recommendation has been addressed, and all five optional suggestions have been implemented as well. Page, table, figure and equation numbers below refer to the revised manuscript.

One change is larger than a normal revision and we describe it plainly at the outset, because the reviewers will notice it immediately.

Reviewer 2's third comment asked for robustness checks across rolling windows and for sensitivity analysis. Implementing those checks exposed a defect in our own implementation. The fused signal was being passed into the mean-variance objective in the raw units of the underlying indicators; on-balance volume takes values of order 10^8 to 10^9, whereas annualised variances are of order 10^{-2}. The linear term therefore dominated the quadratic risk term by roughly ten orders of magnitude, the risk model had no influence on the solution, and the optimiser returned a corner solution holding a single asset on almost every rebalance date. This is precisely the behaviour Reviewer 2 identified in comment 4 as inconsistent with the described risk-budgeting approach. The reviewer was right, and the cause was a defect rather than a design choice.

A second defect surfaced from the same investigation. The regime-dependent volatility targets were specified as absolute levels of 10 to 20 per cent annualised. On a diversified portfolio those levels are never reached, so the constraint bound in one rebalance out of 224 and the regime layer — the central mechanism of the paper — had no effect on the portfolio at all.

We corrected both defects: expected returns are now scaled to a return metric (Equation (4), p. 10) and the regime risk budget is expressed as a fraction of the portfolio's own volatility rather than as an absolute target (Equation (7), p. 11). Following Reviewer 2's optional comment 4, we also extended the evaluation from the original five-year, ten-equity study to an eighteen-year, ten-instrument multi-asset study covering four major stress episodes. The evaluation window runs to every session available at the date the analysis was run, 26 August 2026, rather than stopping at the original submission date. This is not a favourable window chosen after the fact: the 2008–2023 block reproduces exactly the figures the corrected pipeline produced before the extension, and the 665 sessions from January 2024 onward postdate every specification decision in the study. We therefore report them separately, in a new Section 4.2, as an out-of-sample holdout. On that holdout the framework records a higher Sharpe ratio and a smaller maximum drawdown than all three benchmarks, including the two that beat it on Sharpe ratio before 2024.

The consequence is that all reported numbers have changed, and the conclusions are narrower than in the submitted version. The framework's drawdown and stress-period results survive and are now supported across 18 of 18 specifications; the claim of significant alpha does not survive and has been removed; and the signal fusion layer is now reported as a tested null, since ablation shows it contributes nothing. We judged that reporting corrected results with weaker claims was the only defensible course, and we would rather present this to the reviewers now than have it discovered later.

Reviewer 1

Changes which must be made before publication

R1.1 Please fix your literature review. A typical academic paper should have around 45 academic references with about 10 relevant references published in ACADEMIC

JOURNALS in 2025 and 2026. Remove all self-citations and limit them to no more than 2–3 references. ... no more than three citations should be attributed to the same journal or author. In addition, please avoid citing references in clusters of more than three at a time.

Response. The reference list has been rebuilt from the ground up. It now contains **49 references**, of which **27 were published in 2025 or 2026**, well beyond the ten requested. Every entry was resolved against CrossRef, so authors, titles, volume, issue, pages and DOI are taken from the publisher record rather than from memory.

- **Journal concentration.** The submitted version drew 24 of its 33 references from a single journal. No journal now supplies more than three: the maximum of three is reached by The Journal of Finance, The Journal of Portfolio Management, IEEE Access and this journal. Where a further article from this journal was worth citing, it replaced a less relevant one rather than being added on top, so the limit of three is respected.
- **Author concentration.** No author appears more than twice.
- **Self-citations.** There are none, and there were none in the submitted version.
- **Citation clusters.** Every citation command in the manuscript was checked programmatically; none groups more than three references.
- **Scholarly sources.** All 49 entries are journal articles, apart from one standard monograph (Grinold and Kahn, 2000) cited for the information-ratio scaling used in Equation (4). There are no conference proceedings, magazines or trade publications.
- **IJADS references.** Following the Editor's request, three relevant articles from this journal are now cited: Khodamoradi et al. (2023) on robust mean-conditional-value-at-risk portfolio optimisation, Zhang et al. (2025) on the gap between in-sample and out-of-sample model selection, and Mazzotta and Baidoo (2025) on the informational value of alternative data.

Where: References, pp. 28–31; new citations throughout Section 2, pp. 4–7.

R1.2 The paper has some serious acronym-related problems. ... You need to expand an acronym once in the abstract upon the first use and once in the body upon the first use.

Response. The abstract has been rewritten and now contains no unexpanded acronym. In the body, every acronym is expanded at first use: compound annual growth rate (CAGR) and maximum drawdown (Max DD) in Section 3.8; the Akaike and Bayesian information criteria (AIC, BIC) in Section 3.4; information coefficient (IC) in Section 3.5; percentage points (pp) in the caption to Table 6. The five technical indicators are defined in full in Table 2 (p. 9), which is their first appearance anywhere in the manuscript. We also reduced acronym use overall, writing "maximum drawdown" in running text and reserving the abbreviation for table headings.

Where: Abstract, p. 1; Sections 3.4, 3.5 and 3.8, pp. 10–12; Table 2, p. 9.

R1.3 Please carefully polish and clean up all figures and tables. All figures and tables in a paper must be consistent. Please make sure all figures are high-resolution (300 dpi).

Response. All eight figures have been regenerated at exactly 300 dpi from the corrected pipeline, in a single consistent visual style: one typeface, one colour palette, matched line weights, matched axis treatment and matched figure widths. The framework diagram (Figure 1) was redrawn from scratch, both for resolution and because the submitted version depicted a news and text ingestion path that the implementation does not contain.

Two further faults in the submitted version have been corrected. All seven figures previously shared the same `\label`, so every cross-reference resolved to the same figure; each figure now carries a unique label and is referenced individually in the text. All tables have been rebuilt to a single format.

Where: Figures 1–8, pp. 8, 14, 17, 20, 22; Tables 1–13 throughout.

Reviewer 2

Changes which must be made before publication

R2.1 The methodology should be explained more clearly so that other researchers can understand and reproduce the work.

Response. Section 3 has been rewritten with reproduction as its explicit objective, and now runs from p. 7 to p. 12. It states the study universe instrument by instrument, the data source and adjustment, the download window and the resulting evaluation period; gives the exact formula and window for all nine input features in a single table; specifies the mixture model and its refit protocol; states the fusion rule and the confidence term as equations; gives the optimisation problem with every constraint; and sets out the walk-forward protocol and the evaluation design. Nothing in the framework is now described only in prose.

Where: Section 3, pp. 7–12.

R2.2 More implementation details are needed, particularly regarding the data sources used, sampling frequency, sentiment analysis model, feature extraction process, Gaussian Mixture Model settings, and the optimization solver applied.

Response. Each item is now specified explicitly.

- **Data source and sampling frequency.** Daily OHLCV series from Yahoo Finance through the `yfinance` interface, adjusted for splits and dividends; download window 1 January 2007 to 27 August 2026; evaluation period January 2008 to August 2026, comprising 4,689 sessions, of which the final 665 are held out (Section 3.1, p. 7).
- **Sentiment model.** See R2.o3 below. In short: there is none, and the manuscript now says so.
- **Feature extraction.** Table 2 (p. 9) gives the closed-form definition and rolling window of all nine features, followed by the cross-sectional standardisation in Equation (1) and the two composite definitions.
- **Gaussian mixture settings.** Three components, full covariance matrices, three random restarts, fixed seed, fitted on two standardised features, refitted at every rebalance on an expanding window with a minimum of 252 observations (Section 3.4, p. 10).
- **Solver.** CVXPY with the Clarabel interior-point solver; covariance by Ledoit–Wolf shrinkage on 252 trailing sessions (Section 3.6, p. 11).

We also documented one implementation detail that is easy to miss and that we ourselves initially handled incorrectly: mixture component indices permute freely between refits, so labels must be anchored — here by sorting components on fitted volatility — or the regime series is silently scrambled (Section 3.4, p. 10).

Where: Section 3, pp. 7–12; Table 2, p. 9.

R2.3 The experimental results also need stronger statistical support ... statistical significance tests, confidence intervals, robustness checks using different rolling windows, turnover statistics, and sensitivity analysis.

Response. All five are now reported, and this comment is what led to the corrections described in the summary above.

- **Significance tests.** Paired t-test and a Newey–West t-statistic with ten lags on daily excess returns (Table 13, p. 23).
- **Confidence intervals.** A stationary block bootstrap with 21-session blocks and 2,000 replications gives interval estimates for the Sharpe ratio of the strategy, of the benchmark, and of their difference (Table 13, p. 23).
- **Robustness across windows.** A full sweep of three random seeds $\times$ two covariance estimation windows $\times$ three rebalancing intervals. All eighteen cells are reported, not a summary (Table 5, p. 16).
- **Turnover statistics.** Annual one-way turnover, turnover per rebalance, mean and minimum effective number of assets, mean largest weight, and mean and minimum invested exposure (Table 11, p. 21).
- **Sensitivity analysis.** Transaction costs varied from 0 to 25 basis points (Table 11, p. 21), the regime risk budget varied over five specifications (Table 9, p. 19), and component ablations at matched exposure (Table 8, p. 18).

We draw the reviewer's attention to what this analysis shows. The Sharpe improvement over the benchmark is *not* statistically significant: its bootstrap interval is $[-0.102, 0.574]$ and includes zero. Section 4.11 (p. 23) says so directly, and the abstract now states it too. What the evidence supports is consistency rather than significance — 18 of 18 specifications, 18 of 19 calendar years, 4 of 4 stress episodes, and a holdout period the specification never saw — and the manuscript is careful not to conflate the two.

Where: Sections 4.3, 4.6, 4.7, 4.9 and 4.11; Tables 5, 8, 9, 11 and 13.

R2.4 The portfolio allocation results require further clarification. In Figure 5, the portfolio appears concentrated in only a few assets during certain periods, which does not fully align with the described risk-budgeting approach. This behaviour should be explained more clearly.

Response. The reviewer identified a real defect, and we are grateful for it. The concentration was not a property of the strategy but an artefact of unit mismatch in the objective function, described in the summary above. In the submitted version the portfolio held exactly one asset on nearly every date — an effective number of assets of 1.00, with a single holding persisting for 1,364 of 1,421 sessions — because the raw indicator scale overwhelmed the risk term by roughly ten orders of magnitude.

Equation (4) (p. 10) now scales the fused score to a return metric before it enters the objective, which resolves the mismatch. The corrected allocation is diversified and stable: the effective number of assets averages 5.40 and never falls below 4.60, and annual one-way turnover is 0.96 times portfolio value. Section 4.9 (pp. 20–22) explains the failure mode explicitly rather than quietly replacing the figure, since the diagnosis is itself a useful contribution.

Where: Equation (4), p. 10; Section 4.9, pp. 20–22; Table 11, p. 21; Figures 5 and 6, pp. 21–22.

R2.5 The regime detection section also needs stronger justification. The choice of three market regimes should be supported using model selection measures such as

AIC, BIC, or regime persistence statistics.

Response. Table 12 (p. 23) now reports AIC and BIC for two through six components, together with the persistence statistics for the three-state solution: the full transition matrix, self-transition probabilities, mean run lengths, in-sample and out-of-sample state shares, and the annualised volatility and momentum characterising each state.

We report an outcome that does not support our choice, and say so. Both AIC and BIC fall monotonically across the range tested, so neither criterion selects three components; this is the expected behaviour of a mixture fitted to persistent, heavy-tailed financial features, which absorbs tail observations into additional components indefinitely. The choice of three is therefore justified on persistence and interpretability instead, and Table 12 supplies the grounds: the three states are economically distinct (annualised volatility of 12.0, 20.3 and 49.4 per cent), they persist (self-transition probabilities of 0.928 to 0.964, mean run lengths of 14 to 28 sessions), and the transition matrix is close to tridiagonal, so the volatility ordering recovered by the model is economically meaningful. Finer partitions produce states too short-lived for a monthly rebalancing schedule to act on.

Where: Section 4.10, pp. 22–24; Table 12, p. 23; Figure 7, p. 24.

R2.6 In addition, the language and presentation require careful revision. There are grammatical inconsistencies, sudden changes in writing style, and some unfinished formatting elements, including incomplete AI declaration text.

Response. The manuscript has been rewritten end to end in a single voice, so the style discontinuities are gone. Two specific faults the reviewer will have noticed are also fixed: a stray sentence of reviewer-style commentary that had been left inside Section 3.1.1 of the submitted version, and the misspelling of "LLMs" as "LLPs" in Section 2.2.

The AI declaration placeholder has been replaced. We have also added a data and code availability statement.

Where: Throughout; declarations on p. 29.

R2.7 Finally, the equations should be explained in greater detail, with clear definitions of all parameters and constraints used in the optimization model.

Response. Section 3.6 (pp. 11–12) now presents the optimisation as two explicit steps and defines every symbol. The objective terms and every constraint are given, along with the values used and the reason for each: risk aversion $\lambda = 2$; position cap $w_{\max} = 0.25$; turnover penalty $\kappa = c \cdot 252/h$ with $c = 10$ basis points and $h = 21$ sessions, annualised so that trading cost is on the same basis as return and risk; the full-investment equality that makes the first step purely relative; and the regime multiplier k_r that sets total risk in the second step. Equations (1) through (7) each carry an explanation of what the expression does and why it takes that form, including why Ledoit–Wolf shrinkage is required and why the risk budget is relative rather than absolute.

Where: Section 3.6, pp. 11–12; Equations (1)–(7), pp. 9–12.

Suggestions which would improve the quality of the article

All five optional suggestions have been implemented.

R2.o1 The literature review can be strengthened by adding sharper critical comparison between the proposed framework and recent portfolio allocation models.

Response. Section 2.1 now compares the framework against mean–variance optimisation, naive diversification, shrinkage-based optimisation, risk parity and hierarchical risk parity, volatility management, regime-switching allocation and learning-based allocation. Table 1 (p. 5) sets out each paradigm's basis of allocation, its principal limitation and where this study sits, with our own limitation stated in the same terms as everyone else's.

Where: Section 2, pp. 4–7; Table 1, p. 5.

R2.o2 Figures would benefit from more technical interpretation, especially regime transitions, signal confidence behaviour, allocation stability.

Response. Every figure is now interpreted in the text rather than merely described. Regime transitions are analysed through the transition matrix and run lengths (Section 4.10), including the finding that direct calm-to-stress transitions have estimated probability zero. Allocation stability is quantified — effective number of assets, largest weight, turnover — and a new figure (Figure 6, p. 22) plots invested exposure with the transitional and stress states shaded, so the de-risking mechanism is directly visible. Signal behaviour is shown in Figure 8 (p. 24) and interpreted together with the information coefficients in Table 10.

Where: Sections 4.1, 4.5, 4.9 and 4.10; Figures 2–8.

R2.o3 The sentiment-analysis section may be improved by explicitly stating whether domain-specific transformer models or lexicon-based methods were used.

Response. Neither, and the revised manuscript states this plainly in a subsection written for the purpose (Section 3.3, p. 9). No news article, social media post, corporate announcement or any other text is read at any point in the study, and no lexicon, classifier or language model is applied. What the submitted version described as a sentiment pipeline is in fact a composite of four price and volume features, now correctly named the market-state composite and defined in Table 2.

We went further than disclosure. Section 4.8 (pp. 19–20) measures the predictive content of that composite directly and reports that it has none: its information coefficient against next-session cross-sectional returns is -0.026 ($t = -2.26$), and the regime-conditional weighting hypothesis — that market-state information becomes relatively more informative under stress — is not supported: the calm-to-stress contrast is insignificant ($t = -1.46$, $p = 0.145$) with point estimates running the other way. Splitting at the holdout boundary also shows that the negative coefficient is era-specific and disappears after 2023, so the layer cannot be rescued by inverting its sign either. The corresponding portfolio-level ablation shows that removing the layer entirely does not harm performance.

Where: Section 3.3, p. 9; Section 4.8, pp. 19–20; Table 10, p. 20; Table 8, p. 18.

R2.o4 Future work may include cross-market validation, larger asset universes, reinforcement learning-based adaptive parameter tuning.

Response. Two of the three are now in the paper rather than deferred. The evaluation has moved to a ten-instrument multi-asset universe spanning US, developed and emerging equity, government and corporate fixed income, gold and real estate, over eighteen years (Section 3.1, p. 7), and the original mega-capitalisation equity universe is retained as a secondary study in Section 4.12 (p. 25). That secondary result is unfavourable and is reported as such: the framework loses on risk-adjusted return in a concentrated single-sector universe while retaining its drawdown advantage, which identifies the condition under which the framework is useful.

Reinforcement-learning parameter tuning is retained as future work, now stated concretely as learning the risk-budget multipliers, with the overfitting caveats that apply (Section 5.5, p. 28).

Where: Section 3.1, p. 7; Section 4.12, p. 25; Table 14, p. 25; Section 5.5, p. 28.

R2.o5 The manuscript would benefit from minor restructuring to reduce repetition in Sections 4.2 and 4.3.

Response. Those two sections have been merged. Section 4 is now organised so that each subsection answers a distinct question — headline performance, out-of-sample holdout, robustness, stress behaviour, annual consistency, ablation, budget form, signal content, allocation behaviour, regime diagnostics, significance, secondary universe — with no overlapping material. Interpretation that previously appeared twice has been consolidated into Section 5.

Where: Sections 4 and 5, pp. 13–28.

Changes not requested by the reviewers

For completeness, three changes were made that no reviewer asked for.

- **Title.** "Sentiment" has been removed. Once Section 3.3 states that no text or language model is used, the previous title asserted something the methodology contradicts.
- **Abstract.** The claim of "significant alpha" has been removed, because the corrected results do not support it. The abstract now states the return sacrifice and the absence of statistical significance alongside the positive findings.
- **Contributions.** The list has been rewritten around what the evidence supports: the two-step separation of composition from total risk, the matched-exposure ablation design, the relative risk budget, and two documented failure modes.

We believe the revised manuscript is a substantially more careful piece of work than the version submitted, and we thank the reviewers for the comments that forced it.